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Section: 27A

[Assignment-4]

[Answer to the question no 1]

a)

Latch: Level-triggered (changes output as long as input is active)

Flip-flop: Edge-triggered (changes output only on clock edge)

Flip-flops are more stable and used in synchronous systems.

b) JK flip-flop is an improved SR flip-flop

SR FF Truth Table:

S	R	Q
0	0	No change/memory
0	1	0
1	0	1
1	1	Not used

JK FF Truth Table:

J	K	Q
0	0	No change/memory
0	1	0
1	0	1
1	1	Toggle



It overcomes the undefined state of an SR FF ($S=R=1$) because the JK FF

toggles instead of producing an invalid output, giving a desired useful behaviour.

9)

Importance of clock signal in Flip-flops:

1) synchronizes all flip-flops

2) Ensures changes happen at specific times (edges)

✓ ~~3)~~ When the same clock drives multiple flip-flops, the circuit becomes synchronous, avoiding race conditions and providing predictable timing.

2)

State diagrams give a clear graphical view of all states, transitions, and outputs of a sequential circuit.

They simplify understanding, analysis and design, and help verify that the required sequence is correctly implemented.

[Answer to the question no 2]

$$2 \leftrightarrow 4 \leftrightarrow 5 \leftrightarrow 6 \leftrightarrow 7 \leftrightarrow 2$$

$$\begin{array}{l} m = 0 \text{ (up)} \\ m = 1 \text{ (down)} \end{array}$$

M	A	B	C	A ⁺	B ⁺	C ⁺	S _A	R _A	S _B	R _B	S _C	R _C
0	0	0	0	x	x	x	x	x	x	x	x	x
0	0	0	1	x	x	x	x	x	x	x	x	x
0	0	1	0	1	0	0	1	0	0	1	0	x
0	0	1	1	x	x	x	x	x	x	x	x	x
0	1	0	0	1	0	0	x	0	0	x	1	0
0	1	0	1	1	1	0	x	0	1	0	0	1
0	1	1	0	1	1	1	x	0	x	0	1	0
0	1	1	1	0	1	0	0	1	x	0	0	1
1	0	0	0	x	x	x	x	x	x	x	x	x
1	0	0	1	x	x	x	x	x	x	x	x	x
1	0	1	0	1	1	1	1	0	x	0	1	0
1	0	1	1	x	x	x	x	x	x	x	x	x
1	1	0	0	0	1	0	0	1	1	0	0	x
1	1	0	1	1	0	0	x	0	0	x	0	1
1	1	1	0	1	0	1	x	0	0	1	1	0
1	1	1	1	1	1	0	x	0	x	0	0	1

S_A , BC

MA	$B'C'$	$B'C$	BC	BC'
$M'A'$	x	x	x	1
$M'A$	x	x	0	x
MA	0	x	x	x
MA'	x	x	x	1

$$S_A = A'$$

R_A , BC

MA	$B'C'$	$B'C$	BC	BC'
$M'A'$	x	x	1	
$M'A$				
MA	1			
MA'	x	x	x	

$$R_A = MB'C' + M'BC$$

S_B , BC

MA	$B'C'$	$B'C$	BC	BC'
$M'A'$	x	x	x	0
$M'A$	0	1	x	x
MA	1	0	x	0
MA'	x	x	x	x

$$S_B = M'C + MB'C'$$

R_B , BC

MA	$B'C'$	$B'C$	BC	BC'
$M'A'$	x	x	x	1
$M'A$	x			
MA		x		1
MA'	x	x	x	

$$R_B = M'A' + MA'BC'$$

S_C , BC

MA	$B'C'$	$B'C$	BC	BC'
$M'A'$	x	x	x	0
$M'A$	1	0	0	1
MA	0	0	0	1
MA'	x	x	x	1

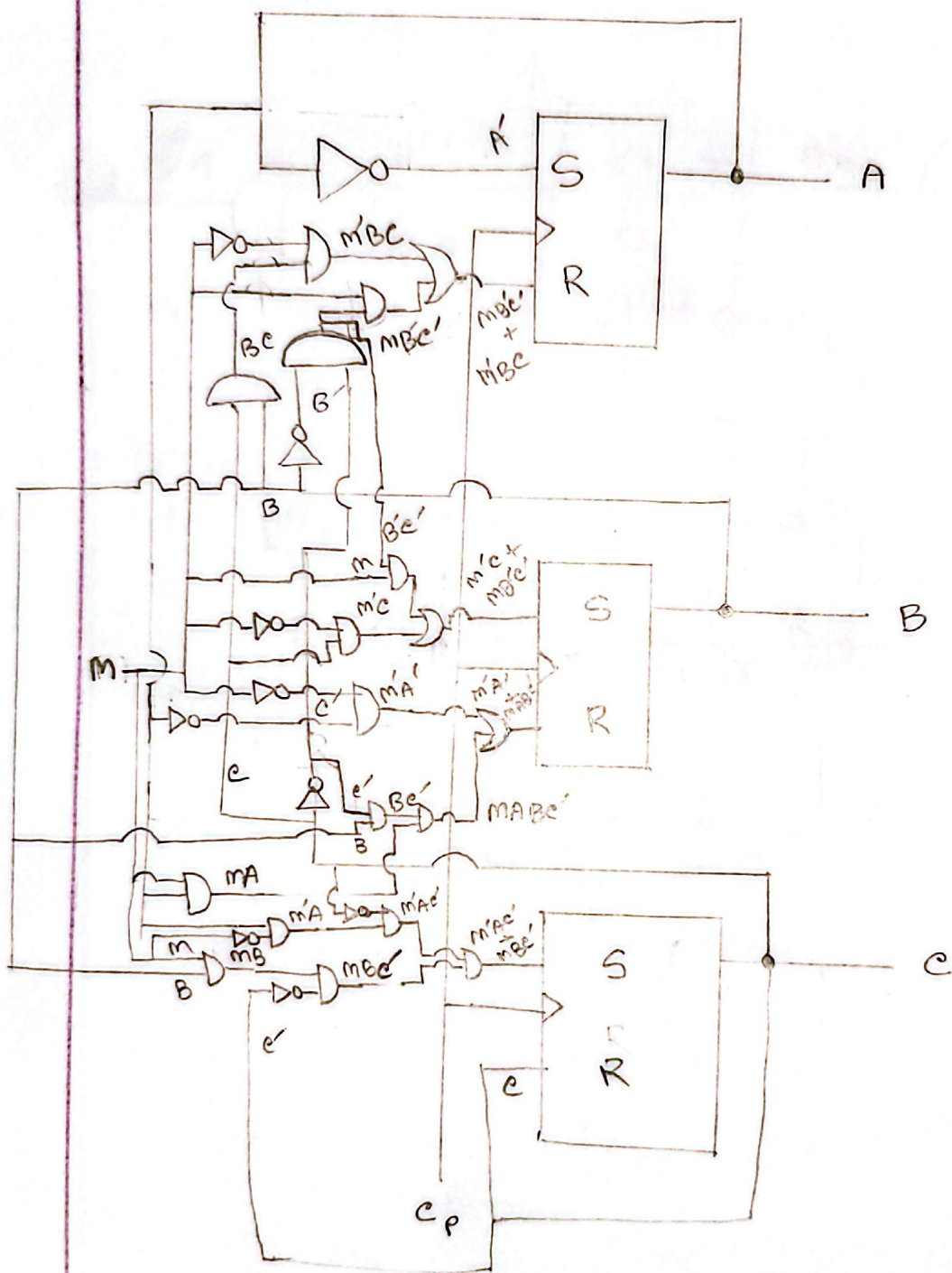
$$S_C = M'AC' + M'BC'$$

R_C , BC

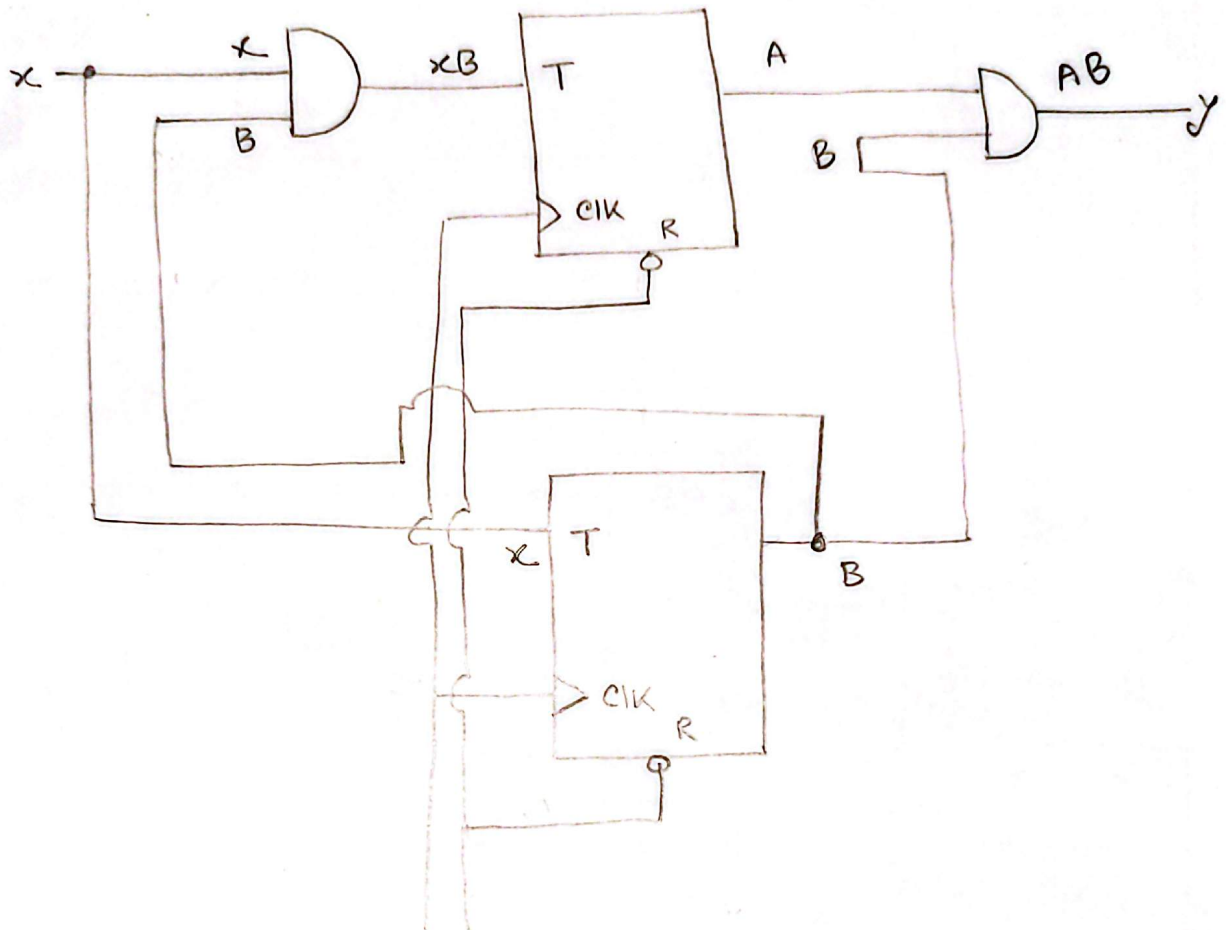
MA	$B'C'$	$B'C$	BC	BC'
$M'A'$	x	x	x	x
$M'A$	0	1	1	0
MA	x	1	1	0
MA'	x	x	x	0

$$R_C = C$$

[P or not loop out of end]



[Ans to the question no 3]

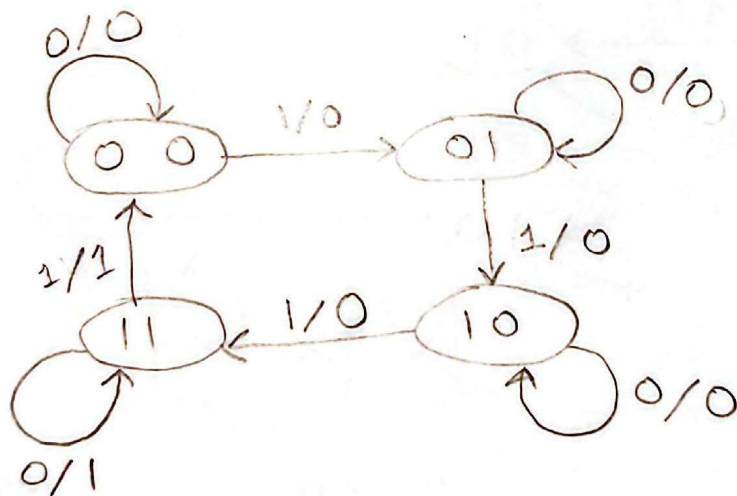


$$TA = xB$$

$$TB = x$$

$$y = AB$$

A	B	x	TA	TB	y	A ⁺	B ⁺
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	1
0	1	0	0	0	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	0	0	1	0
1	0	1	0	1	0	1	1
1	1	0	0	0	1	1	1
1	1	1	1	1	1	0	0



[illegible]

$J_A,$

xy	AB	$A'B'$	AB'	$A'B$
$x'y'$			X	X
xy	(	X	X	X
xy	(X	X	X	X
$x'y'$	1	X	X	X

$$J_A = x + y$$

$K_A,$

xy	AB	$A'B'$	AB'	$A'B$
$x'y'$	X	X	1	
xy	X	X	1	X
xy	X	X	X	X
$x'y'$	X	X	X	X

$$K_A = B$$

J_B

xy	AB	$A'B'$	AB'	$A'B$
$x'y'$	1	X	X	1
xy		X	X	X
xy	X	X	X	X
$x'y'$	X	X	X	X

$$J_B = y'$$

$K_B,$

xy	AB	$A'B'$	AB'	$A'B$
$x'y'$	X	1		X
xy	X	X	1	X
xy	X	X	X	X
$x'y'$	X	X	X	X

$$K_B = A' + y$$

